ITPEC IT Passport Reviewer

Focused on the April 2025, April 2024, and October 2024 question papers

For these papers: 100 compulsory multiple-choice questions, 120 minutes, one answer per question. Use roughly 72 seconds per question on a first pass; mark hard items and return later.

This guide teaches the recurring ideas, gives quick calculations and original self-check questions, and points to questions in your PDFs. The self-check answers are prepared for this reviewer; the folder contains no answer keys for the 2024 or 2025 papers.

Source key

Code	Paper	How to use it
25S	April 2025 IT Passport questions	Main recent paper; attempt after the notes.
24S	April 2024 IT Passport questions	Second recent paper; compare repeated topics.
24A	October 2024 IT Passport questions	Additional practice and variations.

Question references such as 25S Q3 refer to the numbered question in that paper. The observed question blocks are broadly Q1-45 technical, Q46-65 management, and Q66-100 strategy. Use this as a study map, not as a guaranteed future exam blueprint.

Most useful order of study

Priority	Topics	Representative paper questions
1	Security; databases and networks; algorithms and calculations	25S Q1-9, Q26-45; 24S Q1-9, Q26-45; 24A Q1-9, Q26-45
2	System development, projects, service management, governance	25S Q46-65; 24S Q46-65; 24A Q46-65
3	Business strategy, operations, digital transformation	25S Q66-100; 24S Q66-100; 24A Q66-100

1. Logic, numbers, algorithms, and data structures

Logic: AND is true only if every required input is true; OR is true if at least one is true; NOT reverses a value. For three inputs in a chain of AND boxes, all three must be true (25S Q1; 24S Q2; 24A Q2).

Binary: each place is a power of two. 1011 base 2 = 8+2+1 = 11. 101 base 2 = 5, so 1011 x 101 = 55 = 110111 base 2 (25S Q2). A five-bit pattern has place values 16, 8, 4, 2, 1 (24A Q3).

Probability: multiply independent event probabilities. No 1 in three fair die rolls = (5/6)^3 = 125/216 (25S Q4). When selecting people for distinct roles, decide whether one person can hold both roles before multiplying choices (24S Q4).

Classification: recall = true positives / (true positives + false negatives). In 25S Q3, the quality controller found 10 defective items and the model found 5 of them: recall = 5/10 = 0.50. Precision uses model-positive cases as the denominator: 5/(5+15) = 0.25. Do not swap the two.

Flow tracing: write each variable after every assignment. For loops, list the first two and last iterations. Mod gives the remainder; an even integer has i mod 2 = 0. For tiered pricing, calculate the first block at its own rate, then only the excess at the higher rate (25S Q6-7; 24S Q5-9; 24A Q6-8).

Stack versus queue: stack = LIFO (last in, first out); queue = FIFO (first in, first out). If 8, 1, 6, 3 enter a queue, the first two removed are 8 and 1 (25S Q8).

Euclidean remainder pattern: if a box outputs the second input and the remainder of first divided by second, feed (49, 11) to get (11, 5), then feed (11, 5) to get remainder 1 (25S Q9).

Calculation check: identify units, formula, and denominator before reading options. Estimate the answer first; then inspect the choices.

2. Computer systems, software, and reliability

Term	Remember	Questions
CPU / clock	Executes instructions; clock frequency is one factor in performance, not the entire performance story.	25S Q10-13; 24S Q10-13
Cache / RAM / storage	Cache is small and fast near CPU; RAM is working memory; SSD/HDD is persistent storage.	25S Q11-12; 24A Q12-14
OS / driver / BIOS	BIOS or firmware begins startup; OS manages resources and multitasking; driver connects OS to a device.	25S Q17-18; 24A Q12, Q22
OSS / license	Open-source licensing allows access to source under license terms; it does not mean no rules or guaranteed free support.	25S Q21; 24S Q18-19; 24A Q18, Q21
TCO	Total cost of ownership includes acquisition plus running, maintenance, support, and related costs.	24S Q15
Availability	For independent components in series, multiply availabilities. For redundant identical components, at least one works = 1-(1-p)^n.	25S Q14-16; 24S Q14-16; 24A Q14-16

Example: two independent web servers each available 0.8 provide at least one working server with probability 1-(0.2 x 0.2)=0.96. If a database with availability 0.9 is also required, overall availability is 0.96 x 0.9 = 0.864. The independence assumption matters.

Backup is different from redundancy: redundancy helps service continue during a component failure; backups help restore earlier data after loss, corruption, or deletion. Find the last usable backup and count lost changes in time-based backup questions (25S Q19).

Spreadsheet basics: COUNTIF counts cells meeting a condition; IF chooses between outputs. A relative reference moves when copied; a $-locked row or column stays fixed. Practice 25S Q20, 24S Q20, 24A Q17 and Q20.

3. Databases, networks, and web basics

Relational databases

• **Primary key:** uniquely identifies a row. **Foreign key:** references a key in another table and links records (24A Q28; 25S Q28).

• **Normalization:** separates repeating or dependent facts to reduce duplicate data and update anomalies (25S Q27; 24S Q27).

• **Selection** filters rows; **projection** chooses columns; **join** combines related tables. Check which fields are available and whether duplicates change counts (24S Q26-28; 25S Q26).

• **Transaction:** a logical unit of work. COMMIT makes completed changes permanent; ROLLBACK cancels uncommitted changes (25S Q29; 24S Q29).

Networks and messaging

Item	Job / exam clue	Questions
DNS	Maps a domain name to an IP address.	25S Q31
DHCP	Automatically assigns network settings such as an IP address.	24A Q34
NAT	Translates private and public addresses at a network boundary.	24A Q30
Router / gateway	Connects networks; default gateway is the exit from a local network.	25S Q30; 24A Q31
IPv4 / IPv6	IPv4 is a 32-bit address; IPv6 is 128-bit and supports far more addresses.	25S Q32; 24S Q34
HTTPS / TLS	Encrypts browser-server communication in transit; the URL begins https://.	24S Q38; 24A Q32
To / CC / BCC	BCC recipients are hidden from other recipients; CC recipients are visible.	25S Q33; 24S Q30
VPN	Provides an encrypted tunnel over another network.	24S Q44

Also know URL parts (host, domain, path), packets, cookies, online storage, VoIP, and CSV format (24S Q31-33; 24A Q33; 25S Q34; 24A Q92).

4. Security - the most repeated technical block

Across all three papers, roughly Q35-45 concentrates on practical security. Learn the purpose of each control and the right action in a scenario, not just the term.

Idea	Fast distinction	Practice
CIA	Confidentiality = only authorized viewers; integrity = correct and unaltered; availability = usable when needed.	24A Q41
Threat / vulnerability	Threat is a possible harmful event or actor; vulnerability is a weakness it can exploit.	25S Q37
Malware / phishing	Ransomware encrypts data for extortion; phishing tricks users into giving data or access.	25S Q39; 24S Q41; 24A Q38
Zero-day / behavior	Zero-day exploits a flaw before a fix is available; behavioral detection looks for suspicious actions.	24S Q35, Q42
Authentication	Password = knowledge; token = possession; fingerprint or face = biometric; SSO reuses one sign-in across services.	25S Q40, Q45; 24S Q43
Digital signature	A private key signs; the matching public key verifies origin and integrity. Keep the private key secret.	24A Q39; 24S Q39
Hash / encryption	A hash helps detect changes; encryption makes data unreadable without the key. TLS protects transit.	24S Q39; 24A Q32
Risk treatment	Avoid = stop activity; reduce/mitigate = lower likelihood or impact; transfer/share = shift burden; accept/retain = consciously keep.	25S Q44; 24S Q37

ISMS and PDCA: Plan sets policy and risk treatment, Do implements, Check monitors and evaluates, Act corrects and improves. A question asking what follows monitoring normally points to Act (25S Q43; 24A Q37).

Practical incident response: if antivirus flags malware, stop risky activity, isolate or report according to organizational policy, and follow the incident procedure. Do not open more suspicious files or hide the incident (25S Q41; 24A Q36).

Privacy and device handling: limit access to personal data, use secure disposal rather than simple deletion for confidential media, and protect mobile devices with updates, authentication, and careful app installation (25S Q36, Q38;

24S Q36; 24A Q44-45).

5. Development, projects, services, and governance

Concept	What to recall	Question trail
Requirements to acceptance	Define needs, design, build, integrate/test, have users confirm requirements, operate and maintain.	25S Q47-48; 24A Q46-50
Black-box / white-box	Black-box tests input-output behavior without internal design; white-box examines internal logic.	25S Q46; 24S Q48
Maintenance / refactoring	Maintenance changes a delivered system; refactoring improves internal code structure while preserving behavior.	25S Q48; 24A Q47, Q50
DevOps	Development and operations cooperate, automate delivery, and use feedback to improve service.	25S Q50; 24S Q50
WBS / scope	WBS decomposes all project work hierarchically. Scope defines what is included; change requests need impact review.	25S Q52, Q55-56; 24S Q51, Q54, Q56
Schedule / critical path	Longest dependency path determines the earliest completion. Delay on a path with slack may not delay the project.	25S Q53; 24S Q55
Service desk / SLA	Service desk is a single contact point for user inquiries and incidents; SLA states measurable service commitments.	25S Q59-62; 24A Q58-62
Governance / audit / control	Management establishes IT governance; independent audit evaluates controls; separation of duties helps prevent error or fraud.	25S Q63-65; 24S Q63-65

Useful calculations: all-pairs one-to-one communication channels = n(n-1)/2. For 6 people, 15; for 10, 45; thus 3 times as many channels (25S Q54). Work estimate = people x days only if productivity and work divisibility are assumed; 10 x 15 = 150 person-days, so 8 people need 18.75 days (24A Q52).

Charts: Pareto chart ranks causes from most frequent to least, often with cumulative percentage. Use it to prioritize defect or complaint causes (25S Q51, Q71; 24A Q70).

Continuity: facility management covers the physical environment and equipment. A BCP plans how important operations continue after disruption; test and revise it (25S Q58; 24A Q61, Q66).

6. Strategy and business vocabulary

Framework / term	Exam-ready meaning	Practice
SWOT / 3C	SWOT = internal strengths and weaknesses, external opportunities and threats. 3C = company, customers, competitors.	25S Q79, Q81
BSC / KPI	Balanced scorecard connects financial, customer, process, and learning perspectives. KPI tracks progress toward a goal.	25S Q82; 24S Q84
PPM	Product portfolio: stars, cash cows, question marks, dogs by market growth and share.	24S Q79
CRM / SCM / ERP	CRM manages customer relationships; SCM coordinates supply flow; ERP integrates core enterprise resources and processes.	24S Q81; 25S Q84; 24A Q82, Q84
POS / CAD / MRP	POS records sales at checkout; CAD supports design; MRP plans materials from production needs.	25S Q86, Q88; 24S Q86, Q89
RPA / AI	RPA automates repetitive rules-based computer tasks; AI learns patterns or handles less structured judgments.	24A Q86, Q93; 25S Q87
PoC / MVP / RFP	PoC checks feasibility of an idea; MVP is a minimal usable offering to test demand; RFP asks vendors for a proposal.	25S Q96-97; 24A Q85, Q100
DX / UX / IoT	DX changes business models and experiences with digital tech; UX is the total user experience; IoT connects physical devices and data.	25S Q89, Q100; 24S Q98-99; 24A Q81, Q98

Finance formulas: sales = unit price x quantity; gross profit = sales - cost of sales; operating profit = gross profit - selling, general and administrative expenses. Profit result / profit plan x 100% gives achievement ratio (25S Q67, Q70; 24A Q68). ROE uses net income / shareholders' equity; confirm the requested denominator (24S Q69).

Business and compliance: NDA protects confidential information; corporate governance oversees management; compliance means meeting legal and internal obligations. ISO 9001 concerns quality management, ISO 14001 environmental management, and ISO/IEC 27001 information security management (25S Q73-77; 24S Q73-78; 24A Q73-78).

Other recurring terms: benchmarking compares practices with a reference; value engineering seeks function relative to cost; SDGs are the UN's 17 goals; crowdfunding raises funds from many contributors; augmented reality overlays digital content on real scenery (24S Q82-83; 25S Q91-92, Q98; 24A Q99).

7. Self-check - 24 original practice questions

Choose one answer for each question. These are new review questions, not copies of the papers. Try them without the notes, then use the explanations on the next pages.

1. In an AND chain with three inputs, when is the final output true?

A. Any one true B. Exactly two true C. All three true D. None true

2. What is 1011 base 2 multiplied by 101 base 2?

A. 101111 B. 110111 C. 111000 D. 10011

3. A detector flags 8 of 20 actual defective items. What is recall?

A. 0.20 B. 0.40 C. 0.60 D. 0.80

4. Values 4, 7, 9 enter a queue. What is removed second?

A. 4 B. 7 C. 9 D. Cannot tell

5. Two independent servers each have availability 0.9. At least one must work. What is the availability?

A. 0.81 B. 0.90 C. 0.99 D. 1.00

6. A spreadsheet formula is copied down. Which reference keeps row 2 fixed?

A. A2 B. $A2 C. A$2 D. B3

7. Which database key points to a related row in another table?

A. Foreign key B. Primary key C. Candidate key D. Sort key

8. Which operation chooses only specified columns from a relation?

A. Selection B. Projection C. Join D. Commit

9. Which service translates a website name to an IP address?

A. DHCP B. DNS C. NAT D. VPN

10. Which e-mail field hides its recipients from other recipients?

A. To B. CC C. BCC D. Subject

11. A staff member can read records they should not see. Which security property is most directly harmed?

A. Availability B. Integrity C. Confidentiality D. Reliability

12. An attacker exploits a flaw before a fix is available. What is this commonly called?

A. Zero-day attack B. Backup C. Normalization D. Refactoring

13. Which key should the owner keep secret in a digital-signature key pair?

A. Public key B. Private key C. Certificate D. Hash value

14. A company decides to stop an activity that creates an unacceptable risk. Which treatment is this?

A. Accept B. Transfer C. Avoid D. Reduce

15. A tester knows only inputs and expected outputs, not program internals. What test approach is used?

A. White-box B. Black-box C. Static analysis D. Refactoring

16. What decomposes a project's work into a hierarchy?

A. SLA B. WBS C. DNS D. BSC

17. How many one-to-one communication pairs are there among 8 people?

A. 16 B. 24 C. 28 D. 64

18. Where should users normally report an IT service incident?

A. Service desk B. Audit committee C. Supplier board D. Marketing office

19. Which phase of PDCA evaluates measured results against the plan?

A. Plan B. Do C. Check D. Act

20. Which analysis separates internal strengths and weaknesses from external opportunities and threats?

A. 3C B. SWOT C. PPM D. WBS

21. Which system focuses on relationships and interactions with customers?

A. CRM B. MRP C. DNS D. CAD

22. What is the main purpose of a proof of concept?

A. Sign an NDA B. Verify feasibility C. Manage inventory D. Count complaints

23. An employee copies data from forms between systems by following fixed rules every day. Best fit?

A. CAD B. RPA C. IoT D. PPM

24. A proposal request sent to potential vendors is a(n):

A. RFP B. SLA C. BCP D. KPI

8. Self-check answer key and reasons

No.	Answer	Why
1	C	AND requires every input true.
2	B	11 x 5 = 55 decimal = 110111 binary.
3	B	Recall = 8 / 20 = 0.40.
4	B	A queue is FIFO: 4 first, 7 second.
5	C	1 - (1 - 0.9)^2 = 0.99.
6	C	The dollar sign before 2 locks the row.
7	A	A foreign key refers to a key in another table.
8	B	Projection chooses columns; selection chooses rows.
9	B	DNS resolves names to IP addresses.
10	C	BCC addresses are hidden from other recipients.
11	C	Unauthorized reading is a confidentiality breach.
12	A	Zero-day refers to exploitation before a fix is available.
13	B	A signature is created using the private key.
14	C	Stopping the risky activity avoids the risk.
15	B	Black-box testing checks observed inputs and outputs.
16	B	A work breakdown structure is a hierarchy of project work.
17	C	8 x 7 / 2 = 28 pairs.
18	A	A service desk is the user contact point.
19	C	Check compares monitored results with the plan.
20	B	SWOT uses internal S/W and external O/T.
21	A	CRM supports customer relationship management.
22	B	A PoC tests whether an idea works in practice.
23	B	RPA fits repeatable rules-based screen work.
24	A	An RFP requests proposals from vendors.

9. Seven-day finish plan

Day	Work
1	Read sections 1-2; do 25S Q1-25. Write down every unfamiliar term.
2	Read section 3; do 25S Q26-34 and 24A Q26-34.
3	Read section 4; do 25S Q35-45, 24S Q35-45, and 24A Q35-45.
4	Read section 5; do 25S Q46-65 and 24A Q46-65.
5	Read section 6; do 25S Q66-100. Make flashcards for confusing acronyms.
6	Time one full 2024 paper for 120 minutes; review errors by topic, not just question number.
7	Time the other 2024 paper; redo missed calculations and the 24 self-check questions.

On exam day: answer easy definitions quickly; set up formulas on paper; skip and return to long scenarios. For every incorrect practice item, record the correct concept and why the tempting option was wrong.

Last-minute formula card

What to calculate	Formula / quick test
Recall	True positives / (true positives + false negatives)
Independent events	Multiply their probabilities
At least one of n independent identical units works	1 - (1 - availability)^n
One-to-one communication pairs	n(n-1)/2
Profit-plan achievement	Actual profit / planned profit x 100%
Operating profit	Sales - cost of sales - selling/general/administrative expenses

Mistake log - fill this in after a timed paper

Question	Why I missed it	Correct rule or calculation

Prepared from the three PDFs supplied in this folder: 2025S_IP_Questions.pdf, 2024S_IP_Questions.pdf, and 2024A_IP_Questions.pdf. Question references are study pointers; the explanatory text and self-check items in this guide are independently written.